

Data Analysis and Visualization on Titanic and Student's Performance Datasets – An Exploratory Study

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Abstract

Exploratory Data Analysis (EDA) is all about exploring the data in order to identify any underlying pattern before you try to use it to make a predictive model. It also plays a major role in the data discovery process as it is used to analyze data and to recapitulate their different characteristics, which is displayed efficiently with the help of data visualization methods. This paper aims at identifying errors in the dataset, to understand the existing hidden structure and to identify new ones, to detect points in a dataset that deviates to a greater extent from the collected data (outliers), and also is used to find any relationship or intersection between the variables and constants. Two datasets are used namely 'titanic' and 'student's performance' to perform data analysis and 'data visualization' to depict 'Exploratory Data Analysis' which acts as an important set of tools for recognizing a qualitative understanding. As concerning to titanic's data, the survival and non-survival rate depending on the status of the class is visually depicted, alongside with youngest and oldest to survive to understand the insights. Whilst, in student's data score, are plotted, sorted, and cleaned from highest to lowest or vice-versa for understanding the strongest and weakest domain among the students respectively. The datasets were explored and hence it assisted with identifying patterns, outliers, corrupt data, and discover the relationship between the fields in the dataset.

Keywords

Data visualization, Data analysis, Visualization methods, Exploratory data analysis.

1.Introduction

The first and foremost step in a research study is to perform exploratory data analysis. The main intent of performing exploratory data analysis is to aim at obtaining insights to a maximum extent involving various methods. Precisely, these techniques are put into use before initializing a statistical model or conducting complex analysis [1]. The procedure generally describes the dataset in a visual format for easy acknowledgment and to suggest ideas in the decision-making criteria of entities in the business. Visualization of the data deals with factors of identifying and testing data, to determine tendency and to spot the interdependence between sets [2]. Productive use of visualizing supplies a deal of working with huge datasets, exploring underlying features, intersections, structures, and trends on one side, paying way to new foresight and well-organized

decision plans. Hence, a package of the pliability, relativeness, and knowledge of humans, tied with the extensive capacity of storage and computational power of Data Management is a high score point for its acceptance [3].

Reference [4] aims at providing the methods of EDA that convene ways of chances that supply us a base to make decisions. These analyses range focusing on the variable type that involves both categorical and quantitative data. Even further, it aids us to realize variable relationships, especially through both terse of graphical and mathematical plots and correlations. Reference [5] describes different polls, analyzing the methods of fifth-grade, eighth-grade, and high school student's data visualizations methods. 27 students were examined and enquired to give a thought about a question that is historical whilst a usage of the textbook which involved data visualization. Outcomes from quantitative and qualitative analyses focus on the capacity for data visualizations to improve reasoning [6]. The study of a non-linear

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mathematical model to enquire the impact of healthy sanitation and alertness on the spread of Coronavirus disease (COVID-19) existence is analyzed in reference [6] and it makes assumptions that education on the spread of the disease and measurements for prevention promote changes in the behavior of an individual to promote healthy hygiene, paying way to reduce the number of reproduction and burden of disease. Numerical simulations are employed involving real-life data to aid the outcomes. A theoretical model for the cancellation of booking analysis on slots in intercontinental shipping services is initially displayed. Also, involving a case study between Asia and US west coast on container service is then performed based on the proposal of the model. The cancellation range of voyage depending upon changes in ways of statistic and dynamic, the booking factors, and the cases which influences the cancellation aspects are checked and verified in reference [7].

The multivariate model and techniques of statistics are detailed in reference [8] and it stresses the highlight of data science in education and commercial backgrounds. Reference [9] presents the deep discovery for the behavior of the art of tools for linked data visualization. It focuses to provide datasets in the forms of graphs or information chosen by a user based on interest, with the motive to feature their analysis. The tools of visualization are explained and correlated based on their benefits and facilities. The employment of linear and nonlinear methods to visualize data in a lower dimension in the concept of EDA, the suitable method is the one that showcases the availability of natural clusters hidden in the scatter plots are presented in reference [10]. Reference [11] studies the methods of visualization techniques in multiple data concerning facts such as the arrival of flights, the count of passengers, booths, time, and patterns of seasons. This paper examines the correlation between these airports corresponding to different visualization. This work can move to every airport and can further define the analytical methods to forecast waiting time. The aim of reference [12] is to contribute a summary by initializing a visualization for nominees Academy awards of the years between 1993 and 2017. The outcome is five various dashboards are commenced to display the pattern search for the parameters and the Holt-Winters exponential application for flattening prediction. Reference [13] illustrates EDA analyzing the factors: Nottingham Prognostic Index, the Overall Survival Status, and Relapse Free Status collecting the data from the Metabric Breast Cancer

to discuss the rate of survival and recurrence of the disease amidst various patients age for 5-year and 10-years. The EDA is executed employing the tools of visualization and the recorded observations are visualized enabling appropriate swarm plots and tables.

Data analysis generally deals with inspection of data, the transformation of the dataset into a lower dimension, and modeling it to gain effective description and reasoning and to make the conclusion. It is one of the rising areas in the sector of business and technology. Data visualization works with the representation of data in the form of a graph. It converts the raw data present into an approachable way to visualize and recognize the trends. It uses visual tools like charts, graphs and maps, and other visual methods. A brief note has been provided in this section detailing the implementation of data analysis and data visualization, performed in two provided data sets from Kaggle namely “Titanic” and “Student’s Performance”.

2.Data Analysis

It is prominent for an analyst to know the business deals and problems faced by an organization to explore and draw meaningful insights from the raw data. There are four types of data analysis mentioned below.

Descriptive data analysis: Descriptive analysis is the type of analysis that is used to obtain the summary of the set of data points to satisfy every order of the data. It is categorized as one of the prominent methods of conducting statistical data analysis [14]. The descriptive analysis draws a summary for the split of the data, helps to spot outliers, and helps to identify similarities among variables, allowing to perform statistical analysis to a greater extent. The descriptive use aggregation of data and tasks related to mining to fetch results based on previous data.

Diagnostic analysis: Diagnostic analysis analyses and interprets the data deeply to identify the anomalies that cannot be fully explained by current understanding. It looks into the data for previous underlying patterns and determines the casual relationships between patterns that lead to anomalies. This special type of analysis helps to understand future events.

Predictive analysis: The predictive model employs statistics and the solutions are forecasted based on future outcomes. It works with optimization

algorithms to reach possible solutions on factors to attain finer outcomes in the future [15]. It comes up with better guidance on the chances for better outputs.

Prescriptive analysis: Prescriptive analytics has the top capacity today in the field of analytics that determines in developing the areas via predictive and descriptive analytics. It aids the users to provide effective answers to the existing problems and opts for the best choice among various options [16].

3.Data Visualization and Techniques

It assists to handle big data, presenting comparisons, employing the required type of chart, utilizing different palettes in graphs, and also fits apt for the digital age. Visualization is a basic step in the procedure of scientific discovery, although it was not popular till the late 80's until it was fittingly accepted as a stage in the area of research [17]. Leigh et al. highlighted the factor that visualization supports important 3 motives: (i) brief checking of models in simulation process; (ii) fast pace production of obtained results in simulation; and (iii) easy access to communicate to the end-user [18]. In the field of Data Science, the method of visualization is considered a valuable asset in the process of exploring and presenting the different levels of the result; i.e., at the initial and the last procedure of data analysis.

Visually exploring the data is specifically fruitful when source data is known less and the goals of the analysis are not provided in a detailed manner. In this idea presented, visual data exploration is looked forward to accessing an easy step to solve hypothesis-generation procedures, in which they can be accepted or neglected based on visuals, and also new ones can be introduced [19].

The visualization technique is powerful as it explores the data with presentable and interpretable results. It visualizes phenomena that cannot be observed directly. Various visualization methods include bar chart, cartogram, dot distribution map, bullet graph, scatter plot, area chart, bubble cloud, and so on.

Scatterplots serve as a basic method of visualization. The available tools of flexibility lead to the usage of scatter plots in various contexts of exploration and presentation. Scatterplots illustrate every individual object in the data with a point, placed on dimensions of orthogonal and two continuous points [20]. Scatter plots methods have been employed to a greater extent to present statistical graphics and to expose hidden structures in the multivariate dataset. These plots are

identified as one of the finest, polymorphic, and commonly beneficial methods for displaying pairwise axes of correlation structures and low-dimension patterns of the available data alongside the summary for a large total of data [21].

The bar chart is one of the most common visualizations methods which is used to plot x and y axes in the form of a graph for the data in the fashion of comparison of the numerical components. It can be accessed only on the elucidated dataset and is capable of detailing the basic information [22]. Histograms are among the most widely used data visualization methods. A histogram is normally referred to as a chart that displays numeric data in ranges. It shows the number of times a response or range of responses occurs in a data set.

4.Execution Requirements

Pandas is one of the most extensively used libraries in Data Science. Pandas(pd) is an open-source python library for enabling manipulation and analysis of data. Numpy(np) is also one of the important libraries in python. It is used for working with arrays, linear algebra, matrices, and Fourier transform. Matplotlib(plt) and seaborn(sns) are utilized for graphical representation that serves visual purposes. These libraries are accessed to visualize the data in form of graphs and charts to present comparisons.

From the mentioned libraries, there are a lot of functions associated while execution. Some of them include head() which is used to represent the first five columns of the data. tail() represents the last five columns of the dataset. info() specifies the different data types used in the data set and specifies the order and name of the column, count of null and non-null values, and memory.shape gives the count of the number of rows and columns of the dataset. drop() drops or deletes specific columns in the dataset. size returns the size. .ndim returns the number of dimension array. dropna() drops the null values while fillna() fills null values. drop_duplicates() drops values that are duplicated. There are other functionalities explained along with the result obtained.

5.Implementation and Data Analysis

The functions used in this cell are is.null(),sum() in order to find the sum of null values, 'sort_values(ascending=False)' sorts the values present in data in descending order, 'concat()' performs concatenation along the axis, here 'axis=1' specifies the rows, 'keys' to name the columns of the

output. `count()` is employed in order to determine how many times a particular element appears in a provided list or string of values, function `head()` returns the top five rows whereas `tail()` returns the last five rows. `round()` is an inbuilt function that returns round-off values to the given digit along with the floating-point number, if no digit is provided it rounds off to the nearest integer. Here all these functions are employed to fill missing values in Figure 1.

As missing values are filled in the above cell, `'is.null().sum()'` is used to find if there are any null values

```
In [10]: #Missing Data
total = titanic_data_df.isnull().sum().sort_values(ascending=False) #sorts in descending
percent_1 = titanic_data_df.isnull().sum()/titanic_data_df.isnull().count()*100
percent_2 = (round(percent_1, 1)).sort_values(ascending=False)
missing_data = pd.concat([total, percent_2], axis=1, keys=['Total', '%'])
#concatenating total and percent_2 to find the missing data and setting the legend as 'total' and '%'
missing_data.head(5)
```

Out[10]:

	Total	%
body	1188	90.8
cabin	1014	77.5
boat	823	62.9
home_dest	564	43.1
age	263	20.1

Figure 1 Fill missing values

```
In [12]: df.isnull().sum() #checks if there are any missing values
```

Out[12]:

gender	0
race/ethnicity	0
parental level of education	0
lunch	0
test preparation course	0
math score	0
reading score	0
writing score	0
dtype: int64	

Figure 2 Sum of Null values

```
In [12]: #correlation of all columns using pearson method
titanic_cleaned.corr(method='pearson')
```

Out[12]:

	pclass	survived	sex	age	sibsp	parch	fare
pclass	1.000000	-0.312469	-0.124617	-0.408106	0.060832	0.018322	-0.558629
survived	-0.312469	1.000000	0.528693	-0.055513	-0.027825	0.082660	0.244265
sex	-0.124617	0.528693	1.000000	-0.063646	0.109609	0.213125	0.185523
age	-0.408106	-0.055513	-0.063646	1.000000	-0.243699	-0.150917	0.178739
sibsp	0.060832	-0.027825	0.109609	-0.243699	1.000000	0.373587	0.160238
parch	0.018322	0.082660	0.213125	-0.150917	0.373587	1.000000	0.221539
fare	-0.558629	0.244265	0.185523	0.178739	0.160238	0.221539	1.000000

Figure 3 Correlation

`'unique()'` returns the count of unique values, here the count of unique values in the sex field are returned along with its datatype. The mention three functions, `min()`, `max()`, and `mean()` are usually employed to count maximum, minimum, and average values. In Fig. 4, to find the survival and death rates for the youngest and oldest, value '0' is assigned as the initial value for survival and '1' is assigned for the death rate.

present in the dataset. The output obtained is 0 which indicates there are no null values in Figure 2. In Figure 3, `corr()` is employed to perform a pairwise correlation of existing columns in the dataset. Any null values are automatically dropped. It only performs in numeric columns, `'method=pearson'` is used to obtain standard correlation co-efficient. In Pearson method, only uses 3 numerical values where 0 is for no correlation, 1 for positive correlation in total, and -1 for the negative correlation.

The function `min()` is used for youngest as age is less when compared for the oldest, which is calculated using `max()` function and finally `'format'` is used to handle complex strings formatting process in a more efficient way. In Fig. 5, the minimum, maximum and average age for the oldest and youngest on-board is determined to understand the patterns lying in the dataset more efficiently.

```

In [13]: #find unique values
titanic_data_df['sex'].unique()

Out[13]: array([1, 0], dtype=int64)

In [14]: youngest_survive = titanic_cleaned['age'][(titanic_cleaned['survived'] == 1)].min()
youngest_die = titanic_cleaned['age'][(titanic_cleaned['survived'] == 0)].min()
oldest_survive = titanic_cleaned['age'][(titanic_cleaned['survived'] == 1)].max()
oldest_die = titanic_cleaned['age'][(titanic_cleaned['survived'] == 0)].max()
print( "Youngest to survive: {} \nYoungest to die: {} \nOldest to survive: {} \nOldest to die: {}".format(youngest_survive, youngest_die, oldest_survive, oldest_die))

Youngest to survive: 0.1667
Youngest to die: 0.3333
Oldest to survive: 80.0
Oldest to die: 74.0

```

Figure 4 Survival and Death

```

In [19]: print('Oldest person Survived was of:',titanic_data_df['age'].max())
print('Youngest person Survived was of:',titanic_data_df['age'].min())
print('Average person Survived was of:',titanic_data_df['age'].mean())

Oldest person Survived was of: 80.0
Youngest person Survived was of: 0.1667
Average person Survived was of: 29.881134512428055

```

Figure 5 Minimum, Maximum and Average age

‘describe()’ displays the number of values present in the column, count of unique values, frequency, and the top/repeated value in the column, fillna() fills null values for the field accessed. In Figure 6, describe() is used to access information about the embark column in the dataset and the result shows the most frequently used value is ‘s’, so the same value is being employed to fill the null values in the same column.

‘groupby()’ function is used to split the data into groups based on a given condition and ‘replace()’ is an inbuilt function where all occurrences of a substring are replaced with another or new string and returns a copy of it. In Figure 7, the initial column values are being replaced and ‘inplace=True’ is used to modify the data in place as it returns nothing and the data frame is being updated. Followed by that grouping is done based on age in an initial column where mean()

is considered.

‘iloc()’ provides the location of the integer based on indexing. In general, the function is used when the label of the index is not numeric or in case if the index label is not provided. In figure 8, ‘:’ symbolizes extracting all rows with index. In figure 9, random.randn() returns a random number based on a standard normal distribution whereas random.rand() returns a random number based on uniform distribution and they belong to numpy library. ‘crosstab()’ builds a cross-tabulation table that can show the frequency with which certain groups of data occur. In figure 10, a ‘crosstab’ plot visual representation is provided for two existing columns in the dataset with different background gradients or styles.

```
In [16]: titanic_data_df['embarked'].describe()

Out[16]: count      1307
         unique        3
         top           S
         freq         914
         Name: embarked, dtype: object

In [17]: #S is the top value so we are replacing our embarked with S
         titanic_data_df['embarked'] = titanic_data_df['embarked'].fillna('S')
```

Figure 6 describe(), fillna()

```
In [22]: #replacing the initials
         titanic_data_df['Initial'].replace(['Mlle', 'Mme', 'Ms', 'Dr', 'Major', 'Lady', 'Countess',
         'Jonkheer', 'Col', 'Rev', 'Capt', 'Sir', 'Don'], ['Miss',
         'Miss', 'Miss', 'Mr', 'Mr', 'Mrs', 'Mrs', 'Other', 'Other', 'Other', 'Mr', 'Mr', 'Mr'], inplace=True)

In [23]: titanic_data_df.groupby('Initial')['age'].mean()

Out[23]: Initial
         Dona      39.000000
         Master  8.682498
         Miss   23.383099
         Mr     31.933066
         Mrs    36.064275
         Other  44.923077
         Name: age, dtype: float64
```

Figure 7 replace(), groupby()

```
In [15]: df.iloc[:]#integer Location based indexing/ selection by position

Out[15]:
```

	gender	race/ethnicity	parental level of education	lunch	test preparation course	math score	reading score	writing score
0	female	group B	bachelor's degree	standard	none	72	72	74
1	female	group C	some college	standard	completed	69	90	88
2	female	group B	master's degree	standard	none	90	95	93
3	male	group A	associate's degree	free/reduced	none	47	57	44
4	male	group C	some college	standard	none	76	78	75
...	...	...	...	...	...	...	...	...
995	female	group E	master's degree	standard	completed	88	99	95
996	male	group C	high school	free/reduced	none	62	55	55
997	female	group C	high school	free/reduced	completed	59	71	65
998	female	group D	some college	standard	completed	68	78	77
999	female	group D	some college	free/reduced	none	77	86	86

1000 rows × 8 columns

Figure 8 iloc()

```
In [17]: np.random.randn() #returns random number on a standard normal distribution

Out[17]: 0.5313774355071226

In [18]: np.random.rand() #returns random number on a uniform distribution

Out[18]: 0.05374888937644062
```

Figure 9 random.randn(), random.rand()

```
In [45]: pd.crosstab(titanic_data_df.sibsp, titanic_data_df.pclass).style.background_gradient('spring_r')

Out[45]:
```

	pclass	1	2	3
sibsp				
0	100	102	511	
1	113	82	124	
2	8	12	22	
3	4	1	15	
4	0	0	22	
5	0	0	6	
8	0	0	9	

```
In [51]: pd.crosstab([titanic_data_df.sibsp],titanic_data_df.survived).style.background_gradient('winter_r')
```

```
Out[51]:
```

	survived	0	1
sibsp	0	582	309
1	156	163	
2	23	19	
3	14	6	
4	19	3	
5	6	0	
8	9	0	

Figure 10 crosstab()

6.Implementation of Data Visualization

'size()' function returns the size of the data frame where 'unstack()' function is used to convert the dataset into an unstacked format or in general to remove an existing stack. In 'heatmap' colors are used to represent the value in matrix, seaborn library. The darker shade represents the common values while the lighter ones represent the least common activity. In figure 11, the dataset is divided based on two existing columns with a return in size of the data frame in an unstacked format, followed by that a heatmap is plotted.

In figure 12, facegrid plot represents a single variable as well as the relationship between multiple variables

individually. Here, 'facegrid' plot is visualized using 'histogram'. 'add_legend()' is used to add key. In histogram numerical data are distributed, where the data are binned along with the bin count. It usually represents 'univariate' dataset. 'catplot' is a representation in seaborn which shows frequencies of the categories of a categorical variable in figure 13. Here, the visual representation for the survival rate of family and residential alone are shown using 'catplot'. 'subplot' represents a visualization of figures and a set of subplots, from matplotlib library. In figure 14, 'subplot', where parameter 'nrows=1', 'ncols=2'. A histogram is plotted with survival rate. 'set_xticks()' is used to set x ticks locations. 'set_title' is used to set title for the axes where font-weight could also be assigned.

```
In [27]: # Group the dataset by Pclass and Survived and then unstack them
group = titanic_data_df.groupby(['pclass', 'survived'])
pclass_survived = group.size().unstack()

# Heatmap - Color encoded 2D representation of data.
sns.heatmap(pclass_survived, annot = True, fmt = "d")
```

```
Out[27]: <matplotlib.axes._subplots.AxesSubplot at 0x1fea2a8ae50>
```

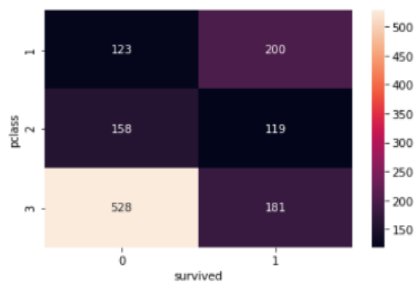


Figure 11 Heatmap


```
In [32]: grid = sns.FacetGrid(titanic_data_df, col='survived', row='pclass', height=2.2, aspect=1.6)
grid.map(plt.hist, 'age', alpha=.5, bins=10)
grid.add_legend();
```

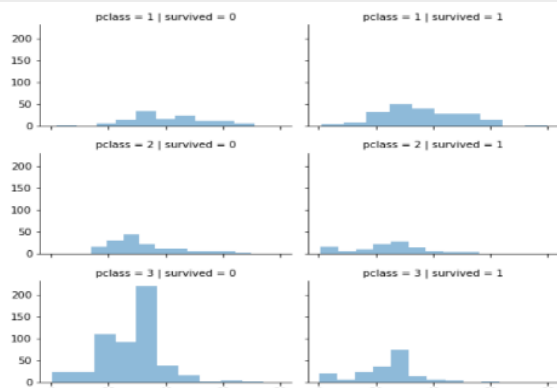
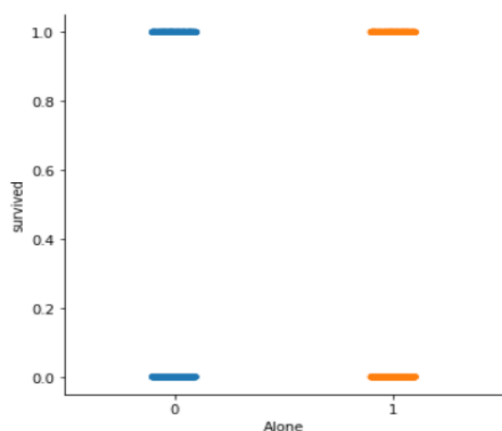


Figure 12 Facegrid in histogram form



```
Out[35]: <seaborn.axisgrid.FacetGrid at 0x1fea2af19d0>
```

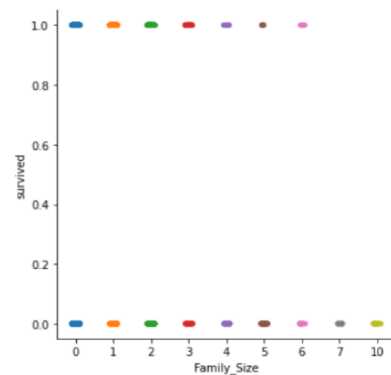


Figure 13 Catplot

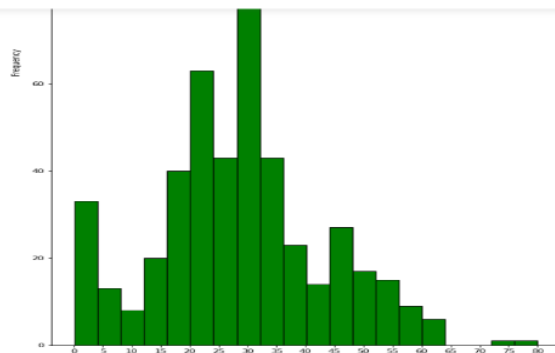
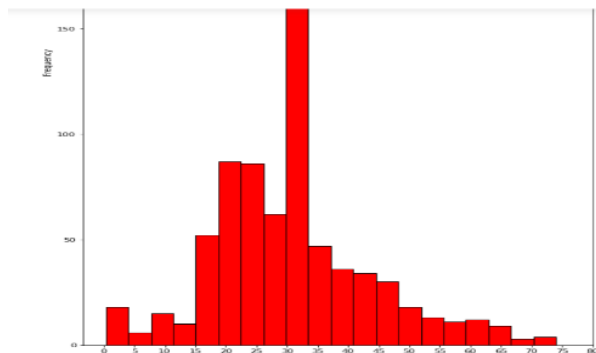


Figure 14 Subplot in histogram form

'distplot' is a feature in seaborn that represents the overall distribution of continuous data variables. In figure 15, a graphical representation of the survival rate for the female and male population has been displayed, using 'distplot()' 'rcParams' in general

defines a runtime configuration for every plot we code

with default style configuration. shows the observation counts in each categorical bin using bars. 'countplot()' shows the observation counts in each categorical bin using bars in figure 16. 'countplot()', parameter 'hue' is allotted for column intake for color encoding, 'x' as

taken as an input and 'palette' as the shade for the plot and the graph is visualized for students writing, reading and mathematics score.

'violin plot' represents many quantitative data across one or more categorical variables in figure 17

'violinplot' is accessed. Here, different columns are passed for parameter 'y', and the width of lines are varied using 'linewidth', and the colors of the plot are given as blue, magma, and red using 'color' parameter.

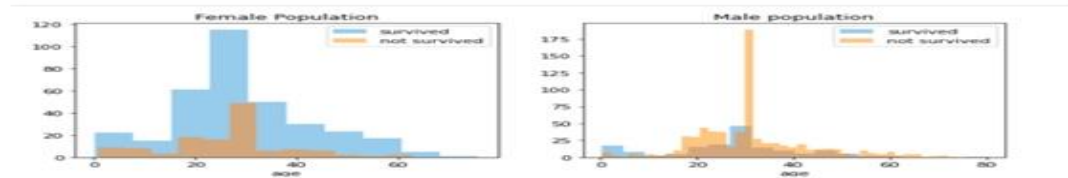


Figure 15 Distplot

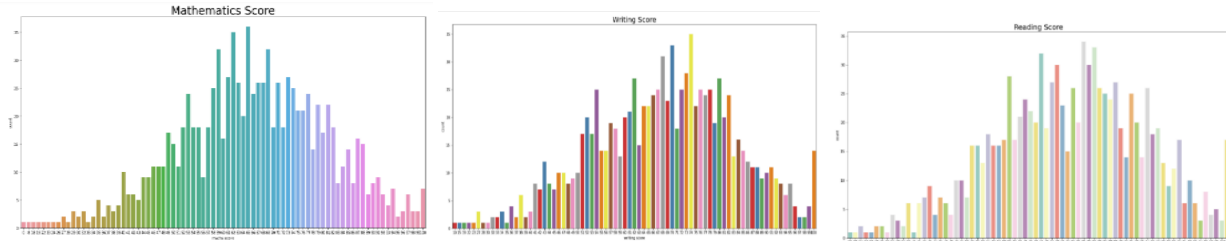


Figure 16 Countplot

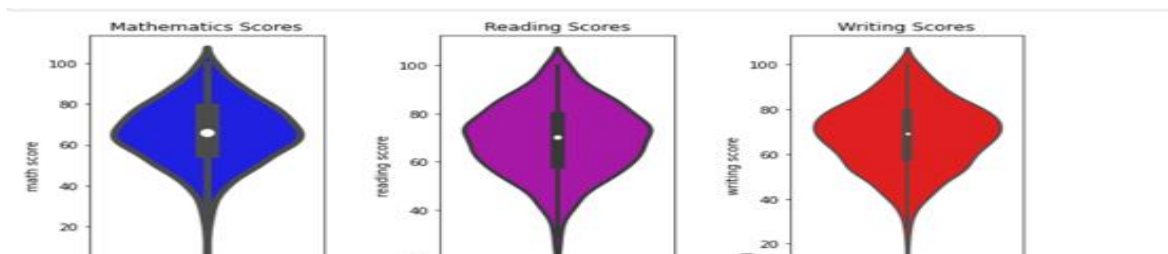


Figure 17 Violin plot

'pyplot' represents a pie-chart where the slices will be ordered and plotted counter-clockwise. In figure 18, a pyplot is plotted for existing columns like student's gender, ethnicity, lunch, and parental level of education exhibiting different styles and variations.

'barplot' displays the central value for a numerical variable using the height of each rectangle. We use 'kwargs' with double stars and it is flexible to handle arguments. **kwargs is a built-in function that takes any number of keyword arguments. 'Keyword

arguments' are declared by a name and if the value is not provided it takes a default value and it will not produce an error.

**kwargs is an 'empty dictionary'. Each undefined argument is stored as key-value pair. In Fig. 19, bar plots accessing different styles and enabling various functions are shown, which are plotted for different columns in the dataset provided with a legend to understand the score in a more precise fashion. In Fig. 20, as already explained 'countplot()' is plotted with different variants.

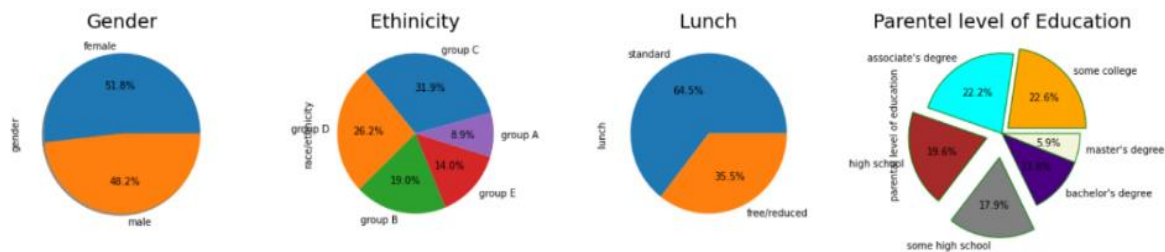


Figure 18 Pyplot

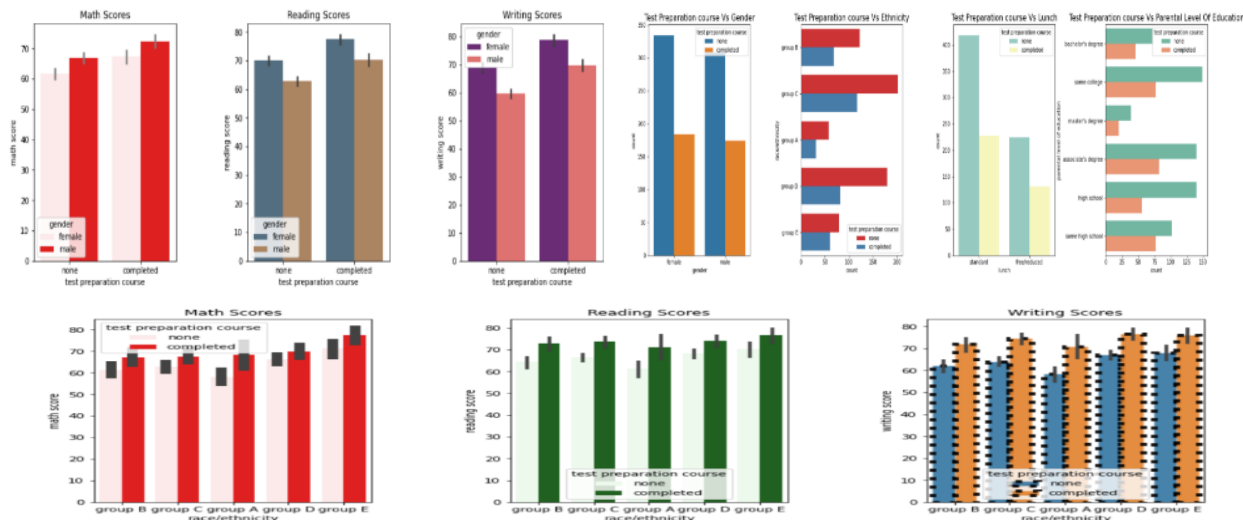


Figure 19 Bar plot

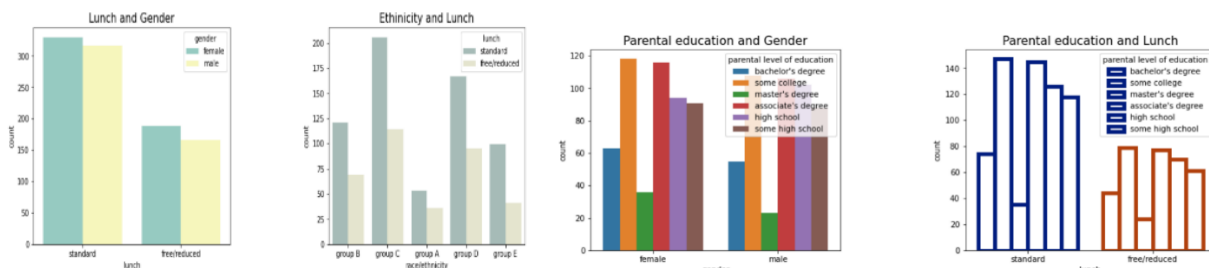


Figure 20 Countplot

7. Discussion

Two datasets are used namely 'titanic' and 'student's performance' to perform data analysis and 'data visualization' to depict 'Exploratory Data Analysis' which acts as an important set of tools for recognizing a qualitative understanding. The datasets were explored and it assisted with identifying patterns, outliers, corrupt data, and discover the relationship between the fields in the dataset.

Visualization reduces the tedious task of reviewing every single existing column to find the common

point, highest and lowest values and it is user-friendly as the graph demonstrates a detailed pattern with easily understandable structures. As concerning analysis, the dataset has been reduced from a higher to a lower dimension, the data is described to know about the existing data types and information about count of values are known, null values are filled using the repeated value, column names have been replaced the way it is recognized by the code performed, grouping and indexing are performed for easy readability. All the mentioned specifications are executed in both datasets. As concerning in titanic's

data, the survival and non-survival rate depending on the status of the class is visually depicted, alongside with youngest and oldest to survive to understand the insights. Whilst, in student's data score, are plotted, sorted, and cleaned from highest to lowest or vice-versa for understanding the strongest and weakest domain among the students respectively.

8. Conclusion

The paper concludes in a way where exploratory data analysis is discussed with understanding the shape of a data set, exploring correlations within the data, and determining if there's a signal for modeling an outcome based on the different features. In conclusion, exploratory data analysis methods are exactly put in before the conventional method of modeling commences and the improvement of sophisticated models in statistics which helps to determine the importance of different features within a data set. Data analysis plays a crucial role as the results are acceptable, identified and detailed correctly, and applicable to required contexts of business. Data analysis techniques have been devised to help in this conclusion. This paper summarizes the main characteristics, often employing visuals. Different techniques which either include non-graphical or graphical and univariate or multivariate are described. This way, it serves the contents required to improve a relevant model for the obstacle at the side to efficiently portray its outcomes.

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Conflicts of interest

The authors have no conflicts of interest to declare.

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